1. **What is a Virtual Machine?**
   A virtual machine is a emulations of a computer. It allows someone to run an operating system within another operating system without messing up the original operating system. Using a Virtual Machine allows the user to work on the other operating system in a separate and isolated environment. Virtual machines can be helpful in various cases and used by various of people. They are often used for testing new software, to run different operating systems, or experimenting with harmful files. The separation between the virtual machine operating system and the computer's original operating system allows users to avoid messing up their own operating system and losing their files.

2. **What is SHA1 and Why Is It Used for File Verification?**
   SHA1, or Secure Hash Algorithm 1, is a cryptographic hash function that creates a 160-bit hash value from an input. This is used for verification often times file integrity. It plays an important role in making sure that files we download from the internet is complete, safe, and unchanged. When the file is downloaded, it makes sure that the computed hash and the hash provided from the source are the same. It allows the user to know that the file has not been tampered with or corrupted. After confirming that the two hash are the same, the user can use the installations safely knowing that it is the correct file.

3. **Virtual Machines Setup and File Verification**
   When the SHA1 has values of the ISO file we downloaded and the one given to us from the website match, we know that the file has not be corrupted, altered, or tampered with. This verification process is extremely important because when the hash is different, it could mean that the file is either incomplete or it isn't safe. When we install a file like that, it could lead to the loss of data, errors in the system, or malware infection. So, it is important to verify the hash before continuing with any installations ensures safe work with the downloaded file. This practice is especially critical for operating system installations, as compromised or altered system files could lead to security vulnerabilities.

4. **Reflection on Installing Windows 10 in VirtualBox**
   Using a virtual machine to download Windows 10 is pretty important when working with files that might cause a security risk, like viruses or untrustworthy downloads. Virtual machines can allow users to create a separate environment to test and interact with unsafe files without putting their computer at risk. If harmful software is downloaded onto the main computer, this may cause the main computer to lose

data or run into problems, or even be hacked. This is why it is important to work on a virtual machine since it allows the user to work in a separate environment on a operating system of their choice. This will ensure that problems met in the virtual machine will not move to the main computer.

5. **Simulated Virus Download and Anti-Virus Scanning**
   After downloading the 10 malware samples in the form of a RAR file and scanning them with anti-virus software, the program detected that 7 out of the 10 files were flagged as malware based on known virus signatures. To further confirm this result, the files were uploaded to VirusTotal, an online malware detection platform that uses multiple anti-virus engines to scan files. VirusTotal confirmed the findings of the local anti-virus software, validating that 7 of the files were indeed malicious. Local anti-virus software is crucial for protecting systems in real-time, as it actively scans for threats and prevents them from causing harm. On the other hand, using an additional tool like VirusTotal provides a second layer of verification by cross-referencing files against numerous malware databases, offering a comprehensive security assessment. Both methods highlight the necessity of scanning files before opening them, as this practice can prevent system infections and data loss.

6. **Reflection on SHA1, Virtual Machines, and Anti-Virus Software**.
   SHA1 hashing, virtual machines, and anti-virus software all work together as a important tool to keep our computer safe. They are the security guards of out computer system. They are important especially when we are dealing with large downloads or suspicious files. SHA1 hashing is a quick way that verify what we downloaded is what we wanted to download. Their unique hash numbers will verify if the file is the same as the one provided by the website. If this number is different, it means that file is either unsafe or incomplete. Virtual machines allows users to work on a operating system in a isolated environment. Allowing users to play with dangerous file that will not cause their host machine to be affected. Lastly, anti-virus software act as a real-time defense. They will detect and remove malware before it can cause damages to your computer.

7. "What Is a Virtual Machine?" *Microsoft Azure*, Microsoft, https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-virtual-machine.
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